Customer & Order Volume Analysis (Northwind SQL Project)

- Objective
  - Analyze customer, order, and product tables to understand data distribution using SQL.
  - Apply COUNT- based and intermediate SQL concepts
  - Identify high – activity regions, suppliers, and products
  - Detect low-activity or inactive customer segments for business insight

- Tools Used
  - SQL Server
  - Northwind Sample Database

Customers by Country

Business Question :

Which countries have the height number of customers?

SQL Concepts Used:

- COUNT
- GROUP BY
- ORDER BY

Key Insights

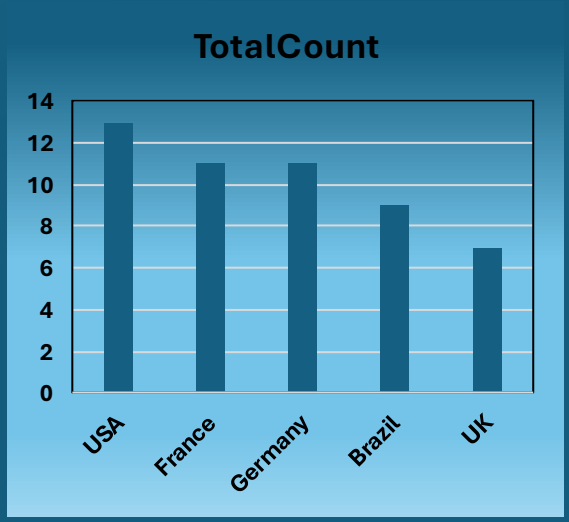
- USA has the highest number of customers (13)
- France and Germany follow with 11 customers each
- Brazil and the UK also show strong customer presence

Business Insight

- Customer distribution is concentrated in a few key countries, indicating primary markets where business activity is highest.

SQL Query:

```
SELECT Country, COUNT(*) AS
TotalCount
FROM Customers
GROUP BY Country
ORDER BY TotalCount DESC;
```



Top Customers — Most Frequently Purchased Products

Business Question

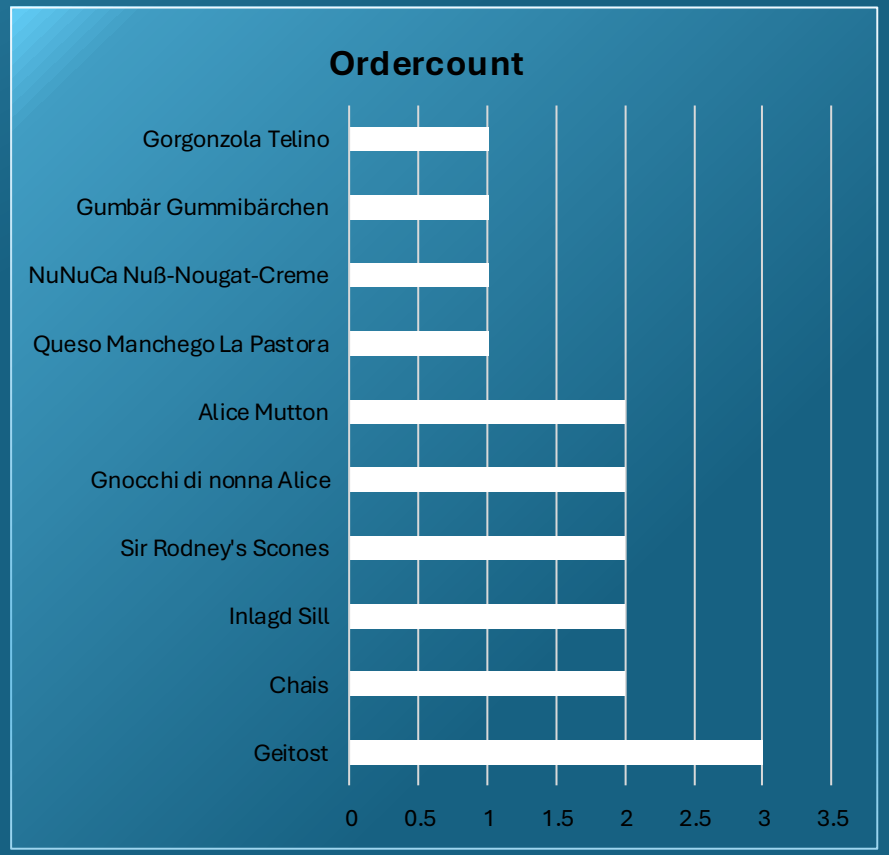
Which products are most frequently purchased by the top repeat customers?

Key Insights

- Top repeat customers tend to purchase specific product multiple times
- Some products appear repeatedly among high- value customers
- Product demand is not evenly distributed across all customers

Business Impact

- Inventory planning can prioritize products frequently purchased by loyal customers
- Personalized promotions can be created based on customer product preferences
- Improves retention by aligning offerings with repeat buying behavior



SQL Logic — Identifying Products Purchased by Top Customers

Analysis Approach

- Identify **top repeat customers** based on order count
- Join customer orders with order details and products
- Count how frequently each product is purchased by **top repeat customers**

SQL Concepts Used

- Subquery
- JOIN
- COUNT
- GROUP BY
- ORDER BY

SQL QUERY

```
SELECT
  p.ProductName,
  tc.CustomerName,
  COUNT(*) AS OrderCount
FROM (
    SELECT TOP 5
      c.CustomerID,
      c.CustomerName,
      COUNT(o.OrderID) AS OrderCount
    FROM Customers c
    JOIN Orders o
      ON c.CustomerID = o.CustomerID
    GROUP BY c.CustomerID, c.CustomerName
    ORDER BY COUNT(o.OrderID) DESC) tc
JOIN Orders o
  ON tc.CustomerID = o.CustomerID
JOIN OrderDetails od
  ON o.OrderID = od.OrderID
JOIN Products p
  ON od.ProductID = p.ProductID
GROUP BY
  p.ProductName,
  tc.CustomerName
ORDER BY
  OrderCount DESC;
```

SQL Logic — Identifying Inactive Customers (Zero Orders)

Analysis Approach

- Analyze customer activity using order counts
- Use a Left Join to retain all Customers
- Identify Customers with no associated orders
- Highlights potential re-engagement opportunities

SQL Concepts Used

- LEFT JOIN
- COUNT
- GROUP BY
- HAVING

Key Insights

- Some registered customers have never placed an order
- Indicates possible issues in onboarding, pricing, or engagement

SQL QUERY

```
SELECT
   c.CustomerID,
   c.CompanyName,
   COUNT(o.OrderID) AS OrderCount
FROM Customers c
LEFT JOIN Orders o
   ON c.CustomerID = o.CustomerID
GROUP BY
   c.CustomerID,
   c.CompanyName
HAVING COUNT(o.OrderID) = 0
ORDER BY c.CompanyName;
```

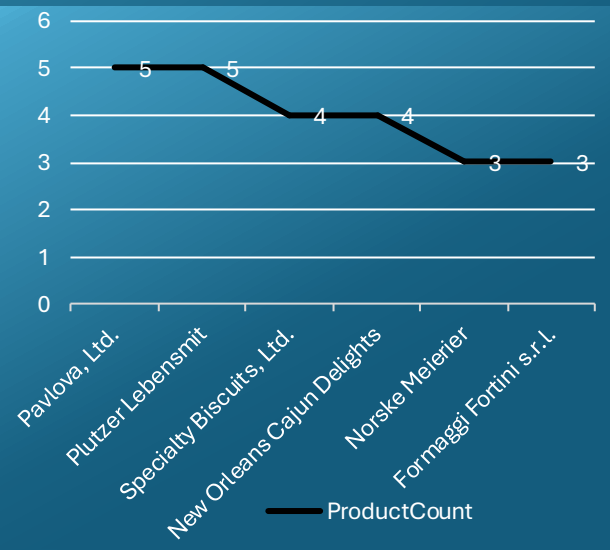
Product Distribution by Supplier

Analysis Approach

- Count number of products supplied by each supplier
- Compare supplier contribution to the product catalog
- Identify concentration of supply

Key Insights

- Pavlova, Ltd. and Plutzer Lebensmit supply the most products (5 each)
- Most suppliers provide between 2–3 products
- Product supply is moderately concentrated among a few suppliers



SQL Query

```
SELECT
s.SupplierName,
COUNT(p.ProductID) AS ProductCount
FROM Suppliers s
LEFT JOIN Products p
ON s.SupplierID = p.SupplierID
GROUP BY s.SupplierName
ORDER BY ProductCount DESC;
```

SQL Logic - Analyzing Order Distribution by Employee

Analysis Approach

- Analyze how orders are distributed across employees
- Count number of orders handled by each employee
- Identify workload imbalance across the sales team

Key Insights

- Order handling is uneven across employees
- Some employees manage significantly more orders
- Indicates potential workload imbalance or role specialization

SQL QUERY

```
SELECT
CONCAT(e.FirstName, ' ', e.LastName) AS EmployeeName,
COUNT(o.OrderID) AS OrderCount
FROM Employees e
LEFT JOIN Orders o
ON e.EmployeeID = o.EmployeeID
GROUP BY
e.FirstName,
e.LastName
ORDER BY OrderCount DESC;
```